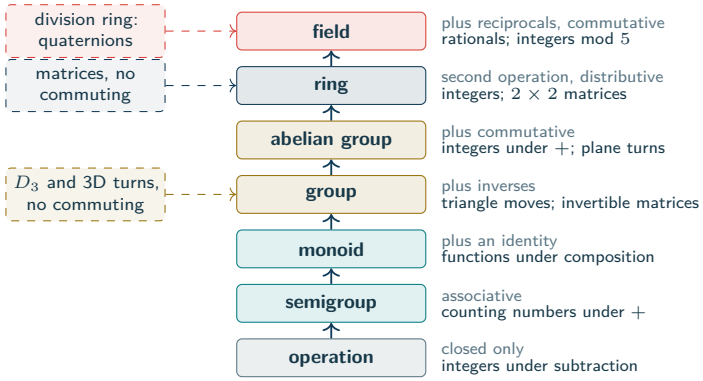
ALGEBRA LADDER 9 OF 9

The map: which laws hold where

Every structure in the ladder is a list of which laws it keeps, and each law buys one thing you can do.

1 / THE RULE AND THE PICTURE

Add one law at a time and a new kind of structure appears



Test a new operation by asking its questions in this order: is it associative, does it have an identity, does every element have an inverse, and does it commute, and with a second operation ask whether it distributes and whether non-zero elements have reciprocals.

laws kept $\Longrightarrow$ name

2 / THE STRUCTURES MET ON THE LADDER

Associativity is almost always kept, and commutativity is the first law to go

structure and operation	associative	commutative	inverses	name
counting numbers, $+$	yes	yes	no	semigroup, no identity
integers, $-$	no	no	no	not even associative
integers, $+$	yes	yes	yes	abelian group
non-zero rationals, $\times$	yes	yes	yes	abelian group
functions, composition	yes	no	yes	monoid; reversible ones form a group
triangle moves D_3	yes	no	yes	group, not abelian
plane turns	yes	yes	yes	abelian group
space turns	yes	no	yes	group, not abelian
2×2 matrices, $\times$	yes	no	only if $\det \neq 0$	monoid; invertible ones form a group
non-zero quaternions, $\times$	yes	no	yes	group, not abelian

3 / WHAT EACH LAW BUYS

A law is a licence to do something

Associativity lets us drop brackets in abc, an identity gives a place to start, and inverses let us solve $a \circ x = b$ by $x = a^{-1} \circ b$. Commutativity lets us reorder freely, and when it fails the side matters: for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ the equation $AX = B$ has solution $X = A^{-1}B = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ while $XA = B$ gives $X = BA^{-1} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$. Distributivity ties two operations together, and reciprocals allow division.

4 / RINGS AND FIELDS IN SMALL WORLDS

Clock arithmetic shows how a field and a ring differ

On a clock with 5 hours every non-zero number has a reciprocal, since $2 \times 3 = 6 \equiv 1$, so it is a field and equations $ax = b$ are always solvable. On a clock with 6 hours $2 \times 3 = 6 \equiv 0$, so two non-zero numbers multiply to zero and 2 has no reciprocal, which makes it a ring only. The 2×2 matrices behave like the 6-hour clock in this respect, with zero products of non-zero matrices, and the quaternions behave like the 5-hour clock except that they do not commute.

5 / CLASSIFY A NEW OPERATION

Define $a \circ b = a + b + ab$ on the integers

It commutes and associates because $a \circ b = (1 + a)(1 + b) - 1$, which is ordinary multiplication seen through the shift $a \mapsto 1 + a$, and it has identity 0 since $a \circ 0 = a$. The inverse of a would be $-\frac{a}{1+a}$, which is an integer only for $a = 0$ and $a = -2$, so the operation forms a commutative monoid, and the shift explains each law at once.

6 / TRY THESE

Name the structure

(a) The six moves of a triangle under composition form a group that is not abelian. **(b)** The integers under $+$ and $\times$ are a commutative ring and not a field, since $\frac{1}{2}$ is missing. **(c)** Check the claim "associativity holds for matrices" with $(RF)S = R(FS)$ from the matrix sheet, and check that commutativity fails with $RF \neq FR$.

ASIDE / THE WHOLE LADDER

The nine sheets climb from addition and multiplication of whole numbers, through composition of actions and matrices, to rotations in space.

LINKS TO OTHER SHEETS

The ladder runs: 1 operations and their laws, 2 distributivity, 3 identity and inverse, 4 letters and proof, 5 composition, 6 triangle moves, 7 matrices, 8 rotations and quaternions, and this map.

Convention. Group: associative with identity and inverses; abelian: also commutative; ring: two operations with distributivity; field: a commutative ring where non-zero elements have reciprocals; D_3 is the symmetry group of the triangle.

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